SUBJECT: ***Project NY ERFO FW WARI 2019-1(1) – Geotechnical Recommendations for Gabion Baskets Retaining Walls in the Wallkill River National Wildlife Service***

TO	MESSAGE/COMMENT	FROM/DATE
G. Riffle Assistant Project Manager K. Roman Project Manager Through: M. Carruthers Project Management Branch Chief	**INTRODUCTION** This Geotechnical Memorandum summarizes the results of our subsurface exploration, laboratory testing, and geotechnical evaluations and recommendations for the construction of two (2) gabion basket retaining walls along the Winding Waters Trail located in Wallkill River National Wildlife Refuge, in Orange County, New York. Figure No. 1 in Appendix A shows the site location and vicinity maps. **GENERAL PROJECT INFORMATION** **Background** Heavy rain and storms caused erosion and slope failures in Winding Water trail at sites 240 and 324. **Project Description** The project includes the construction of two (2), 300 ft long, earth retaining structures at sites 240 and 324, to re-establish embankment of the river. **EFLHD GEOTECHNICAL STUDY** **Site Geology** The geologic map of New York State (Fisher, et al., 1970) indicated that the project sites are located within the Wappinger Group, which consists of limestone, dolostone, shale, and chert of Cambrian-Lower Ordovician age. The geologic map of the project area is included in Figure No. 2 in Appendix A. **PROCEDURES AND RESULTS** **General** A subsurface exploration program was conducted on March 2022 by EFLHD Subsurface Exploration Team. The program consisted of drilling five (5) soil borings, three (3) borings in Site 240 and two (2) borings in Site 324. Borings were drilled using a CME 750 drill rig mounted on a rubber tired all-terrain vehicle (ATV). Boreholes were advanced using 3-¾-inch (ID) hollow stem augers (HSA) and were terminated upon reaching auger refusal or after reaching the planned borings depth. Boring locations were laid out by EFLHD personnel in the field by referencing topographic features, measuring distances from existing surface features, and using a hand-held GPS equipment. Approximate boring locations is shown on the Borings Location Plan in Appendix B.	Y. Perez, Geotechnical Engineer YESENIA STEPHANIE PEREZ SOTO Digitally signed by YESENIA STEPHANIE PEREZ SOTO Date: 2024.09.09 15:07:42 -04'00' M. Abouzakhm, PE Division Geotechnical Engineer MOUNIR A ABOUZAKHM Digitally signed by MOUNIR A ABOUZAKHM Date: 2024.09.10 07:10:29 -04'00' Through: S. Whittemore Technical Services Branch Chief 9/9/2024

Sampling

Soil samples were obtained using a standard split-barrel (split-spoon) sampler driven during Standard Penetration Tests (SPT) in general accordance with ASTM D 1586, Standard Method of Penetration Test and Split Barrel Sampling of Soils. SPT was performed by driving the split-spoon sampler 24 inches into relatively undisturbed soil under the impact of a 140-pound automatic hammer free falling 30 inches. SPT was performed at 2.5-foot intervals. The number of blows required to advance the sampler through each 6-inch interval was recorded on field boring logs. The number of hammer blows required to advance the split-spoon sampler the middle foot is designated as the "Standard Penetration Resistance" or N value. Boreholes were backfilled with auger cuttings.

Representative portions of the split-spoon samples were preserved in glass jars. The samples were visually classified and logged in the field by the drilling crew. Samples were then delivered to the EFLHD Materials Testing facility in Sevierville, TN for testing and storage.

Data Summary

The results of the field tests and measurements were recorded on the driller's logs and appropriate data sheets in the field. These data sheets and logs contain information concerning the boring methods, samples attempted and recovered, indications of materials encountered. They also contain interpretations by the exploration foreman of the subsurface conditions based on the performance of the equipment and cuttings brought to the surface by the drilling tools. Therefore, the field data represents both factual and interpretative information.

The boring logs in Appendix C represent a compilation of field and laboratory data and description of the soil samples by the geotechnical engineer.

LABORATORY TESTING

A laboratory testing program was conducted on representative soil samples recovered during the subsurface explorations to aid in the classification, and the evaluation of engineering parameters of the subsurface materials. The following tests were performed on selected samples.

- Grain Size Distribution (AASHTO T-11 and T-27)
- Moisture Content (MC) (AASHTO T-265)
- Plastic Limit (PL) (AASHTO T-90)
- Liquid Limit (LL) (AASHTO T-89)

The laboratory testing results are presented in Appendix D and are summarized in Table No. 1.

Table No. 1 - Summary of Laboratory Test Results.

Boring No.	Sample No.	Depth (ft)	% Gravel	% Sand	% Fines	MC (%)	PL (%)	LL (%)	PI (%)	AASHTO Classification
BH 22-01	J-3	5-7	0	12.5	87.5	34.4	26.3	36.9	10.6	A-6(11)
BH 22-01	J-5	10-12	1.6	51.8	46.6	21.6	NP	21.6	NP	A-4(0)
BH 22-01	J-6	13-15	0	5.5	94.5	22.1	NP	NV	NP	A-4(0)
BH 22-02	J-4	8-10	-	73.5	24.7	87.0	NP	NV	NP	A-2-4(0)
BH 22-02	J-7	15-17	0	5.3	94.7	23.8	NP	28.8	NP	A-4(1)
BH 22-03	J-2/J-3	3-7	2.5	66.2	31.3	28.7	NP	NV	NP	A-2-4(0)
BH 22-03	J-4/J-5	8-12	5.4	66.9	27.7	17.0	NP	NV	NP	A-2-4(0)
BH 22-04	J-2/J-3	3-7	2.2	54.3	43.5	35.1	NP	NV	NP	A-4(0)
BH 22-04	J-4	8-12	2.9	53.8	43.3	62.9	NP	NV	NP	A-4(0)
BH 22-05	J-2/J-3	3-7	0.4	22.6	77.0	37.0	24.1	37	12.9	A-6(10)
BH 22-05	J-4	8-10	0	57	43.0	-	NP	NV	NP	A-4(0)
BH 22-05	J-6	13-15	0.7	41.7	57.6	18.7	NP	NV	NP	A-4(0)

FINDINGS

Soil Borings

Descriptions of the subsurface conditions encountered in the soil borings are provided below. The stratification lines designating the interfaces between soil types on the boring logs in Appendix C represent approximate boundaries. The transition between materials may be gradual. It should be noted that one or more of the units may be absent at specific locations. See Appendix C for more detail information. The subsurface layers encountered were as follows:

Fill

A 7 ft to 10 ft thick layer of fill was encountered in the soil borings. The fill consisted Primarily of silt, silty sand, and lean clay.

Silty Sand (SM)

Underneath the fill, silty sand (SM) was encountered. This layer extended to depths of 13 ft, 15 ft, 13.3 ft, 12 ft, and 14.9 ft below existing grades in Borings BH22-01, BH22-02, BH22-03, BH22-04, and BH22-05, respectively. N values for this layer ranged from weight of hammer (WH) to 13 blows per foot (bpf), indicating very loose to loose relative densities. Possible boulders may be encountered in this layer.

Silt and Sandy Silt (ML)

Underneath the above listed stratum, a layer of silt was encountered in BH22-01, and BH22-02, and extended to depths of 17.4 ft and 19.42 ft below existing site grades, respectively. A layer of sandy silt was encountered on BH22-05 and extended to a depth of 14.9 ft below existing site grades. N values for this layer ranged from 3 to 21 bpf, indicating firm to hard consistencies.

Groundwater

Groundwater was encountered at depths of 9.3 ft and 10.7 ft in borings BH22-04 and BH22-05, respectively, during the drilling operation. However, fluctuations in groundwater levels should be expected due to seasonal variations, precipitation, evaporation, and surface water runoff.

RECOMMENDATIONS AND ANALYSIS

Recommendations

Based on our observations during the site reconnaissance, the results of our subsurface exploration program, geotechnical analyses, and discussions with our partner, it was concluded that gabion baskets wall with a maximum height of 18 ft and a base of 15 ft is recommended for sites 240 and 324. The wall foundations should be supported on firm natural soils, engineered fill placed on firm natural soils, decomposed rock, or bedrock. In addition, to increase the lateral resistance of the proposed walls, we recommend that the base of the walls be anchored to the foundation soils using a 1-inch diameter, 5 ft long steel dowel every 6 ft along the length of the walls. Bioengineering and riprap revetment were also considered for the subject site. However, given the height of the failed embankment, and the water velocities provided by the Hydraulic team, it was determined that an engineering structure such as gabion basket wall was warranted. Such structures are known to provide long term stability, minimum need for maintenance and minimum disturbance.

Geotechnical Analysis

The slope stability analysis of the proposed gabion baskets were conducted using limit equilibrium methodology, in general accordance with the design concepts and procedures presented in "AASHTO LRFD Bridge Design Specifications, 9th Ed." (2020) design requirements. A uniform distributed live surcharge load of 250 lb/sq ft (psf) was applied along the trail to account for maintenance traffic loading following AASHTO-LRFD, 2020, Section 3.11.6.4 requirements. A summary of the soil engineering parameters assumed in the external, internal, and global stability analyses of Winding Water Trail sites 240 and 324 are presented in Table No. 2.

Table No. 2 - Summary of assumed engineering properties for analysis.

Material Type	Unit Weight (pcf)	Cohesion (psf)	Internal Friction Angle ϕ (°)
Silty Sand (SM)	110 – 120	0.0	24° – 28°
Silt (ML)	115 – 125	0 – 100	24° – 28°
Apparent Bedrock	138 – 140	0 – 800	38° –42°
Granular Backfill	110 – 120	0.0	34° – 36°
Gabion Basket	110 – 120	Infinite Strength	

The selected load factors and combinations are presented in Table No. 3. The factors follow (AASHTO-LRFD, 2020).

Table No. 3 - Retaining wall load factors and combinations.

Load Combination	EH	EV	ES	LS	EQ
Strength I (Max)	1.50	1.35	1.50	1.75	-
Strength I (Min)	0.90	1.00	0.75	1.75	-
Service I	1.00	1.00	1.00	1.00	-
Extreme Event I	1.00	1.00	1.00	1.00 or 0.50	1.00
Extreme Event II	1.00	1.00	1.00	0.5	-

Table No. 4 - Gravity Gabion Wall design resistance factors.

Analysis	Resistance Factor
Sliding	Φ: 0.90
Bearing	Φ: 0.45
Overall/Global	Φ: 0.75 (FS:1.3)
Internal Stability – Joint Resistance	Φ: 0.75
Internal Stability - Overturning	Φ: 1.00
Overall/Global Extreme Event I	Φ: 0.9 (FS:1.1)
Overall/Global Extreme Event II	Φ: 0.9 (FS:1.1)

External and internal stability analysis computations were performed using PTC Mathcad Prime v. 8.0 software program by Computer Software Company (PTC) Inc., and the computer program Slide 2 by Rocscience, Inc. Copies of the computer output are included in Appendix E.

According to AASHTO 11.5.4.2, a seismic design shall not be considered mandatory for walls located in Seismic Zones 1 through 3, or for walls at Sites where the site adjusted peak ground acceleration is less than or equal to 0.4g, unless one or more of the following is true:

- Liquefaction induced lateral spreading or slope failure, or seismically induced slope failure, due to the presence of sensitive clays that lose strength during the seismic shaking, may impact the stability of the wall for the design earthquake.
- The wall supports another structure that is required, based on the applicable design code or specification for the supporting structure, to be designed for seismic loading and poor seismic performance of the wall could impact the seismic performance of that structure.

In addition, according to C11.5.4.2, paragraph 7, a seismic analysis should be considered mandatory if the wall is in seismic zone 2 or higher, and if either of the following is greater than 30 feet:

- The exposed wall height plus the average depth over the width of the wall of any soil surcharge present.
- For tiered walls, the sum of the exposed height of all the tiers plus the average soil surcharge depth.

Based on the above listed criteria, we concluded that the proposed retaining walls do not require an external or internal seismic evaluation due to an adjusted PGA less than 0.4g, and the sites are in Seismic Zones 1 to 3. However, an Extreme Event I global stability analysis is required for all sites. For this analysis, an assumed allowable lateral deformation was established based on the flexible nature of the types of structure.

An Extreme Event II design was performed according to AASHTO 11.5.4 due to flooding in the area.

The proposed retaining walls satisfied the minimum static and Extreme Event I (seismic) and II (flood), and global stability requirements with the factors of safety listed in Table No. 5.

Table No. 5. Global/Overall Stability Calculated Factors of Safety

Site	Calculated Factor of Safety		
	Static	Seismic	Extreme Event II
Winding Water Trail	2.20	1.96	2.91

Details of the global stability analyses are included in Appendix E.

DISCLAIMER/LIMITATIONS CLAUSE

The subsurface explorations and tests described in the section Procedures and Results of this memorandum have been conducted in accordance with standard practices and procedures (except as specifically noted). The results of these exploration and test represent conditions at the specific locations indicated. Subsurface conditions beyond these locations may vary. The analyses recommendations sections of this memorandum include interpretations and recommendations developed by the Government in the process of preparing the design. These interpretations are not intended as a substitute for the personal investigation, independent interpretation, and judgment of the Contractor.

REFERENCES

AASHTO, 2020 (Errata 2021). *AASHTO LRFD Bridge Design Specification.* 9th ed. Washington, D.C.: American Association of State Highway and Transportation Officials.

FHWA, 2014. *Standard Specifications For Construction of Roads and Bridges on Federal Highway Projects,* Washington D.C.: US Department of Transportation.

Fisher, D., Isachsen, Y. & Rickard, L., 1970. *Geologic Map of New York State, consisting of 5 sheets: Niagara, Finger Lakes, Hudson-Mohawk, Adirondack, and Lower Hudson, New York State Museum and Science Service, Map and Chart Series No. 15, scale 1:250,000..* [Online] Available at: https://mrdata.usgs.gov/geology/state/

FLH, 2017. *Rock Characterization Guidelines,* Washington D.C.: US Department of Transportation - FHWA.

FLH, 2017. *Soil Description and Identification Guidelines,* Washington D.C.: US Department of Transportation - FHWA.

APPENDIX:

- Appendix A – Figures
- Appendix B – Boring Location Map
- Appendix C – Boring Logs
- Appendix D – Laboratory Testing Results
- Appendix E – Design Analysis
- Appendix F – Selected Photographs

Appendix A – Figures

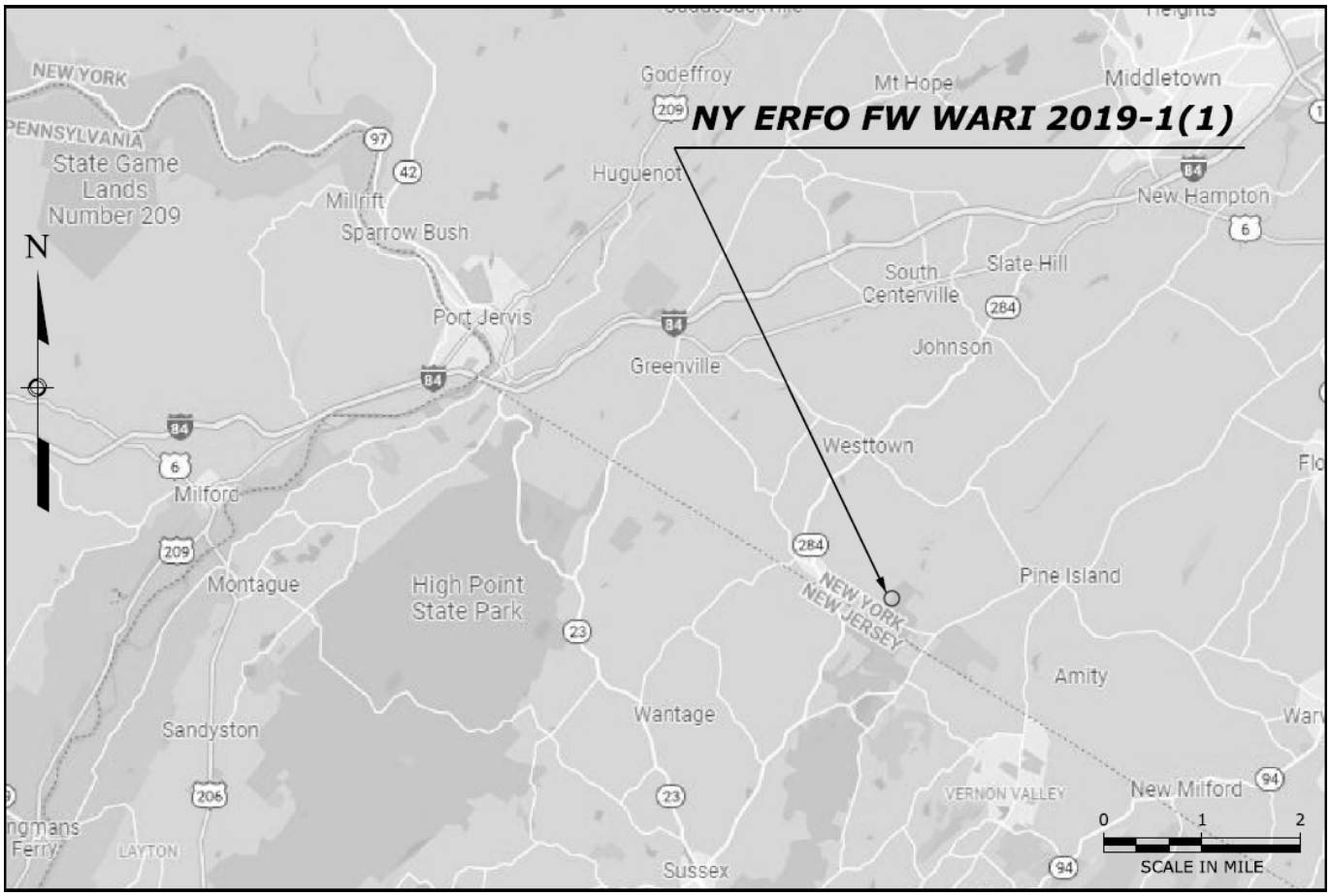


Figure No. 1: Location Map

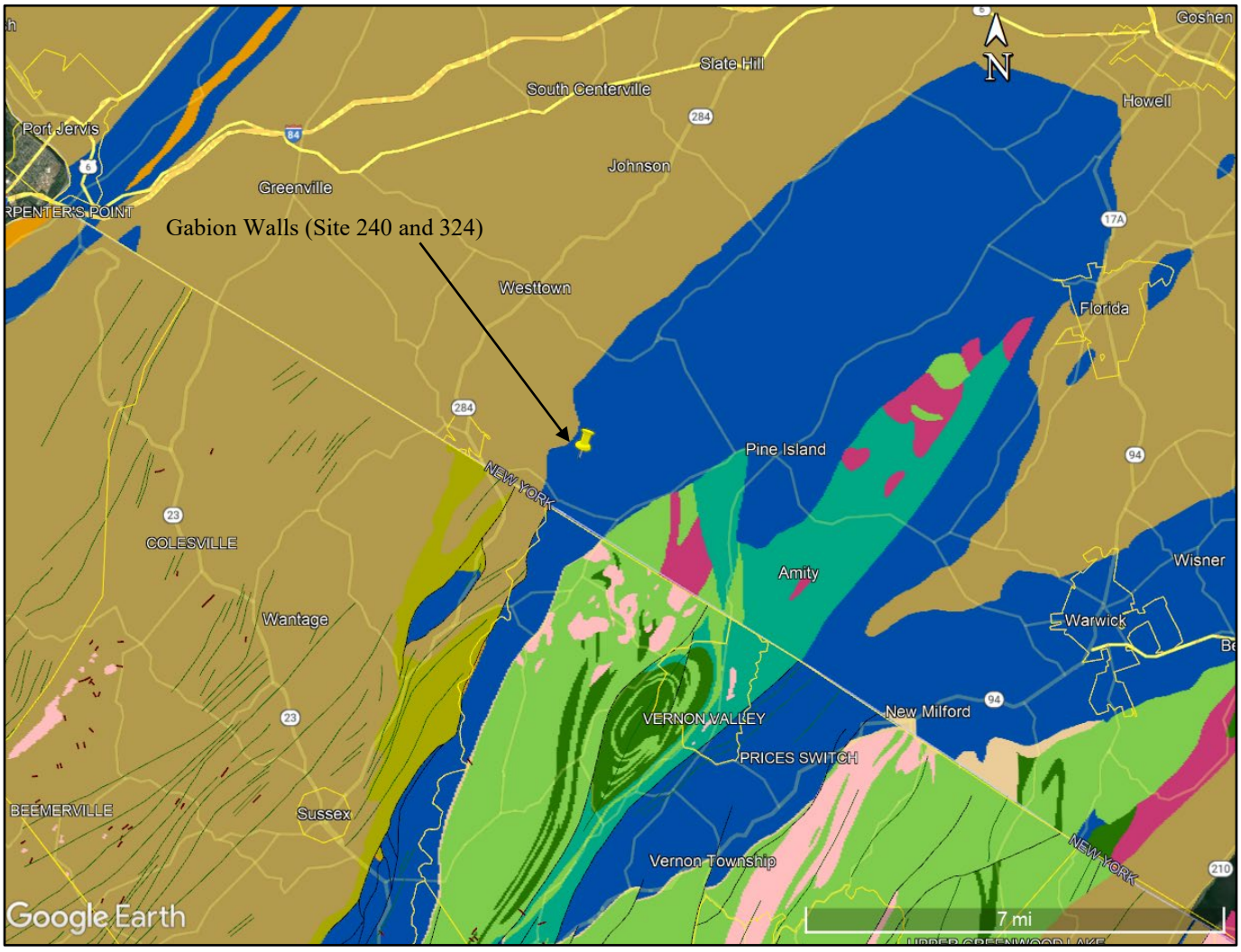


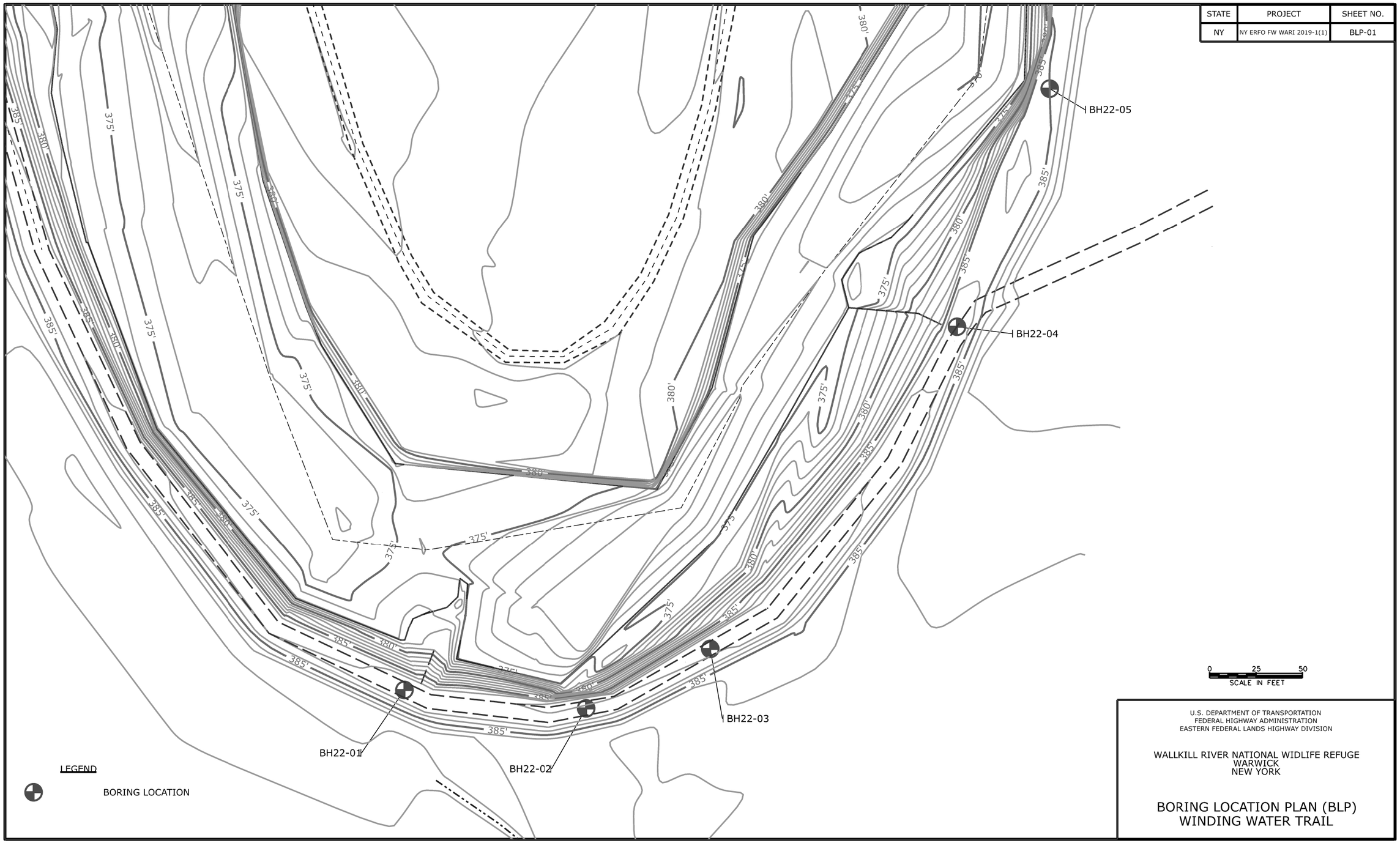
Figure No. 2: Geologic Map of New York State (Fisher, Isachsen, & Rickard, 1970)

Map Legend	
	Wappinger Group *– "(Including Fishkill limestone and dolostone): Copake Formation? - limestone, dolostone; Rochdale Formation-limestone, dolostone; Halcyon Lake Dolostone-locally cherty; Briarcliff Dolostone; Pine Plains Formation-dolostone, shale, oolite; Stissing Formation-dolostone, shale."*

Fisher, D.W., Isachsen, Y.W., and Rickard, L.V., 1970, Geologic Map of New York State, consisting of 5 sheets: Niagara, Finger Lakes, Hudson-Mohawk, Adirondack, and Lower Hudson, New York State Museum and Science Service, Map and Chart Series No. 15, scale 1:250,000.

Appendix B – Borings Location Maps

STATE	PROJECT	SHEET NO.
NY	NY ERFO FW WARI 2019-1(1)	BLP-01



LEGEND

BORING LOCATION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
EASTERN FEDERAL LANDS HIGHWAY DIVISION

WALLKILL RIVER NATIONAL WIDLIFE REFUGE
WARWICK
NEW YORK

BORING LOCATION PLAN (BLP)
WINDING WATER TRAIL

Appendix C – Boring Logs

BORING LOG

US Department of Transportation
Federal Highway Administration
Eastern Federal Lands Highway Divion

Project Name:	NY ERFO FW WARI 2019-1(1)	**Boring No.:**	BH22-01	**Logged By:**	G. Moody
Project Location:	8 State Line Rd, Westtown, NY 10998	**Boring Location:**	41.29498, -74.5268	**Operator:**	J. Farley
Project Datum:	NAD83	**Hammer Wt/Type:**	140 lbs/Automatic	**Weather:**	
Surface Elev.:	387.5 ft	**BH Type/Size:**	3.75	**Boring Began:**	March 20, 2022
Groundwater Depth (ft):		**LeadEngineer**	Y. Perez Soto	**Completed:**	March 20, 2022

Elev.(ft) / Depth(ft)	Material Description	Sample No.	Recovery (%)	PP / RQD	SPT Blows per 6 in	Atterberg Limits LL (%)	Atterberg Limits PL (%)	SPT N Value
387.5	Ground Surface							
0	**Fill** Silt, soft to firm, dark brown, moist.	J-1	0.60		0-1-1-1			2
		J-2	0.9		1-2-3-4			5
	A-6(11)	J-3	1.25	3.0	1-3-4-3	36.9	26.3	7
		J-4	1.25		1-3-3-3			6
377.5								
10	**Silty Sand (SM)** Loose, gray, moist. A-4(0)	J-5	1.65	1	0-1-2-3	21.6		3
374.5								
13	**Silt (ML)** Firm to very stiff, gray moist. A-4(0)	J-6	1.40	3.0	4-5-13-15			18
		J-7	1.40		9-10-11-12			21
370.08								
17.42	Auger Refusal at 17.42 ft.							

Remarks: Boring was backfilled with auger cutting upon completion.

PROJECT NY ERFO FW WARI 2019-1(1)

SHEET: 1 of 1

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BORING LOG

US Department of Transportation
Federal Highway Administration
Eastern Federal Lands Highway Divion

Project Name:	NY ERFO FW WARI 2019-1(1)	**Boring No.:**	BH22-02	**Logged By:**	G. Moody
Project Location:	8 State Line Rd, Westtown, NY 10998	**Boring Location:**	41.29498, -74.52641	**Operator:**	J. Farley
Project Datum:	NAD83	**Hammer Wt/Type:**	140 lbs/Automatic	**Weather:**	
Surface Elev.:	387.8 ft	**BH Type/Size:**	3.75	**Boring Began:**	March 19, 2022
Groundwater Depth (ft):		**LeadEngineer**	Y. Perez Soto	**Completed:**	March 19, 2022

Elev.(ft) / Depth(ft)	Material Description	Sample No.	Recovery (%)	PP / RQD	SPT Blows per 6 in	Atterberg Limits LL (%)	Atterberg Limits PL (%)	SPT N Value
	Ground Surface							
379.8 / 8	**Fill** Silty Sand - Very Loose to loose, dark brown, gray, moist.	J-1	0.55		0-1-1-1			2
		J-2	0.60		2-3-2-2			5
		J-3	0.60		1-2-2-2			4
372.8 / 15	**Silty Sand (SM)** Very loose to loose, gray, moist. A-2-4(0)	J-4	1.60		1-1-2-2			3
		J-5	1.50		1-5-5-4			10
		J-6	1.25	2.0	2-2-7-4			5
368.38 / 19.42	**Silt (ML)** Stiff to hard, gray, moist. A-4(1)	J-7	1.45	3.0	1-4-7-8	28.8		11
			2.5	2.5	4-9-50/5"			50/5"
	Bottom of boring at 19.42 ft.							

Remarks: Boring was backfilled with auger cutting upon completion.

PROJECT NY ERFO FW WARI 2019-1(1)

SHEET: 1 of 1

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BORING LOG

US Department of Transportation
Federal Highway Administration
Eastern Federal Lands Highway Divion

Project Name:	NY ERFO FW WARI 2019-1(1)	**Boring No.:**	BH22-03	**Logged By:**	G. Moody
Project Location:	8 State Line Rd, Westtown, NY 10998	**Boring Location:**	41.2949, -74.52603	**Operator:**	J. Farley
Project Datum:	NAD83	**Hammer Wt/Type:**	140 lbs/Automatic	**Weather:**	
Surface Elev.:	388.5 ft	**BH Type/Size:**	3.75	**Boring Began:**	March 18, 2022
Groundwater Depth (ft):		**LeadEngineer**	Y. Perez Soto	**Completed:**	March 18, 2022

Elev.(ft) / Depth(ft)	Symbol	Material Description	Depth Scale (ft)	Sample No.	Recovery (%)	PP / RQD	SPT Blows per 6 in	Atterberg Limits LL (%)	Atterberg Limits PL (%)	SPT N Value
388.5		Ground Surface								
0		**Fill** Silty Sand (SM) - Very Loose, dark brown, gray, moist. A-2-4(0)	0							
			1	J-1	0.86		1-1-2-2			3
			2							
			3							
			4	J-2	0.65		1-1-1-2			2
			5							
			6	J-3	0.90		1-2-2-3			4
380.5			7							
8		**Silty Sand (SM)** Loose, dark brown, gray, moist. A-2-4(0)	8							
			9	J-4	0.90		4-5-8-4			13
			10							
			11	J-5	0.40		2-7-4-6			11
			12							
375.17			13	J-6			50/4"			50/4"
13.33		Bottom of boring at 13.33 ft.	14							
			15							
			16							
			17							
			18							
			19							
			20							
			21							
			22							
			23							
			24							
			25							
			26							
			27							
			28							
			29							
			30							

SPT N Value: 0 10 20 30 40 50 60 70 80

Remarks: Boring was backfilled with auger cutting upon completion.

PROJECT NY ERFO FW WARI 2019-1(1)

SHEET: 1 of 1

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BORING LOG

US Department of Transportation
Federal Highway Administration
Eastern Federal Lands Highway Divion

Project Name:	NY ERFO FW WARI 2019-1(1)	**Boring No.:**	BH22-04	**Logged By:**	G. Moody
Project Location:	8 State Line Rd, Westtown, NY 10998	**Boring Location:**	41.29557, -74.5257	**Operator:**	J. Farley
Project Datum:	NAD83	**Hammer Wt/Type:**	140 lbs/Automatic	**Weather:**	
Surface Elev.:	386 ft	**BH Type/Size:**	3.75	**Boring Began:**	March 18, 2022
Groundwater Depth (ft):	9.3	**LeadEngineer**	Y. Perez Soto	**Completed:**	March 18, 2022

Elev.(ft) / Depth(ft)	Material Description	Sample No.	Recovery (%)	PP / RQD	SPT Blows per 6 in	Atterberg Limits LL (%)	Atterberg Limits PL (%)	SPT N Value
	Ground Surface							
376 / 0	**Fill** Silty Sand (SM) - Very Loose to Loose, dark brown, gray, moist. A-4(0)	J-1	0.80		1-1-4-6			5
		J-2	0.90		1-2-2-2			4
		J-3	0.70		1-2-2-2			4
	A-4(0)	J-4	1.95		2-2-2-5			4
10	**Silty Sand (SM)** Loose, gray, wet.	J-5	0.65		1-2-6-6			8
373.84 / 12.16	Auger refusal at 12.16 ft.							

18 Mar 2022

Remarks: Boring was backfilled with auger cutting upon completion.

PROJECT NY ERFO FW WARI 2019-1(1)

SHEET: 1 of 1

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BORING LOG

US Department of Transportation
Federal Highway Administration
Eastern Federal Lands Highway Divion

Project Name:	NY ERFO FW WARI 2019-1(1)	**Boring No.:**	BH22-05	**Logged By:**	G. Moody
Project Location:	8 State Line Rd, Westtown, NY 10998	**Boring Location:**	41.29591, -74.52558	**Operator:**	J. Farley
Project Datum:	NAD83	**Hammer Wt/Type:**	140 lbs/Automatic	**Weather:**	
Surface Elev.:	385 ft	**BH Type/Size:**	3.75	**Boring Began:**	March 16, 2022
Groundwater Depth (ft):	10.7	**LeadEngineer**	Y. Perez Soto	**Completed:**	March 16, 2022

Elev.(ft) / Depth(ft)	Material Description	Sample No.	Recovery (%)	PP / RQD	SPT Blows per 6 in	Atterberg Limits LL (%)	Atterberg Limits PL (%)	SPT N Value
Ground Surface								
385 / 0	**Fill** Lean Clay (CL) with sand, very soft to soft, dark brown, gray, moist. A-6(10)	J-1	0.95		WH-WH-1-1			1
		J-2	1.0	2.0	WH-WH-3-2	37	24	3
		J-3	0.80	2.0	1-1-2-1			3
378 / 7	**Silty Sand (SM)** Very loose, gray, moist. A-4(0)	J-4	2.0		WH-WH-1-1			1
		J-5	1.65		WH-WH-2-3			2
373 / 12	**Sandy Silt (ML)** Firm, gray, wet. A-4(0)	J-6	1.2	4.0	1-3-4-50/5"			7
370.08 / 14.92	Auger refusal at 14.92 ft.							

16 Mar 2022

Remarks: Boring was backfilled with auger cutting upon completion.

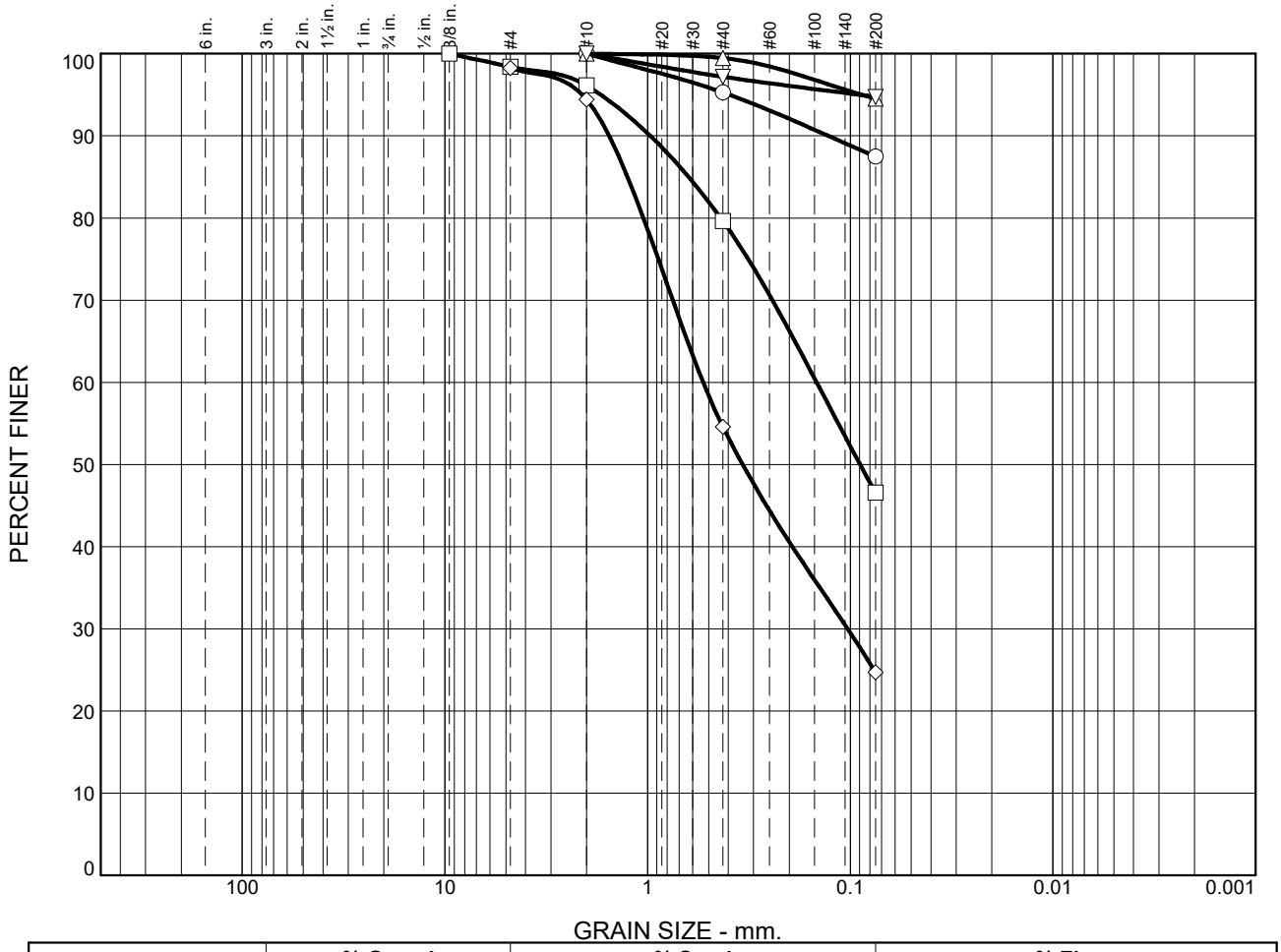
PROJECT NY ERFO FW WARI 2019-1(1)

SHEET: 1 of 1

RSLog / EFL Geotechnical Log -SPT_YPS / fhwa-eastern-federal-lands-highway-div / admin / December 19, 2022 01:22 PM

Appendix D – Laboratory Testing Results

Particle Size Distribution Report



	% +3"	% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
○	0.0	0.0	0.0	0.0	4.7	7.8	87.5	
□	0.0	0.0	1.6	2.3	16.5	33.0	46.6	
△	0.0	0.0	0.0	0.0	0.5	5.0	94.5	
◇				3.8	39.8	29.9	24.7	
▽	0.0	0.0	0.0	0.0	2.8	2.5	94.7	

SOIL DATA

	SOURCE	SAMPLE NO.	DEPTH (ft.)	Material Description	AASHTO
○	W Waters	01/J-3	5-7	silt	A-6(11)
□	W Waters	01/J-5	10-12	silty sand	A-4(0)
△	W Waters	01/J-6	13-15	silt	A-4(0)
◇	W Waters	02/J-4	8-10	silty sand	A-2-4(0)
▽	W Waters	02/J-7	15-17	silt	A-4(1)

FHWA

Sevierville, TN

Client: FHWA/EFLHD

Project: Winding Waters Trail

Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0228-0232

Tested By: CJ/CH **Checked By:** _____

These results are for the exclusive use of the client for whom they were obtained. They apply only to the samples tested and are not indicitive of apparently identical samples.

Particle Size Distribution Report



	% +3"	% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
○	0.0	0.0	2.5	11.7	27.5	27.0	31.3	
□	0.0	0.0	5.4	15.2	29.2	22.5	27.7	
△	0.0	0.0	2.2	3.8	19.1	31.4	43.5	
◇	0.0	0.0	2.9	1.5	24.0	28.3	43.3	
▽	0.0	0.0	0.4	0.2	7.8	14.6	77.0	

SOIL DATA

	SOURCE	SAMPLE NO.	DEPTH (ft.)	Material Description	AASHTO
○	W Waters	03/J-2/J-3	3-7	silty sand	A-2-4(0)
□	W Waters	03/J-4/J-5	8-12	silty sand	A-2-4(0)
△	W Waters	04/J-2/J-3	3-7	silty sand	A-4(0)
◇	W Waters	04/J-4	8-12	silty sand	A-4(0)
▽	W Waters	05/J-2/J-3	3-7	lean clay with sand	A-6(10)

FHWA

Sevierville, TN

Client: FHWA/EFLHD

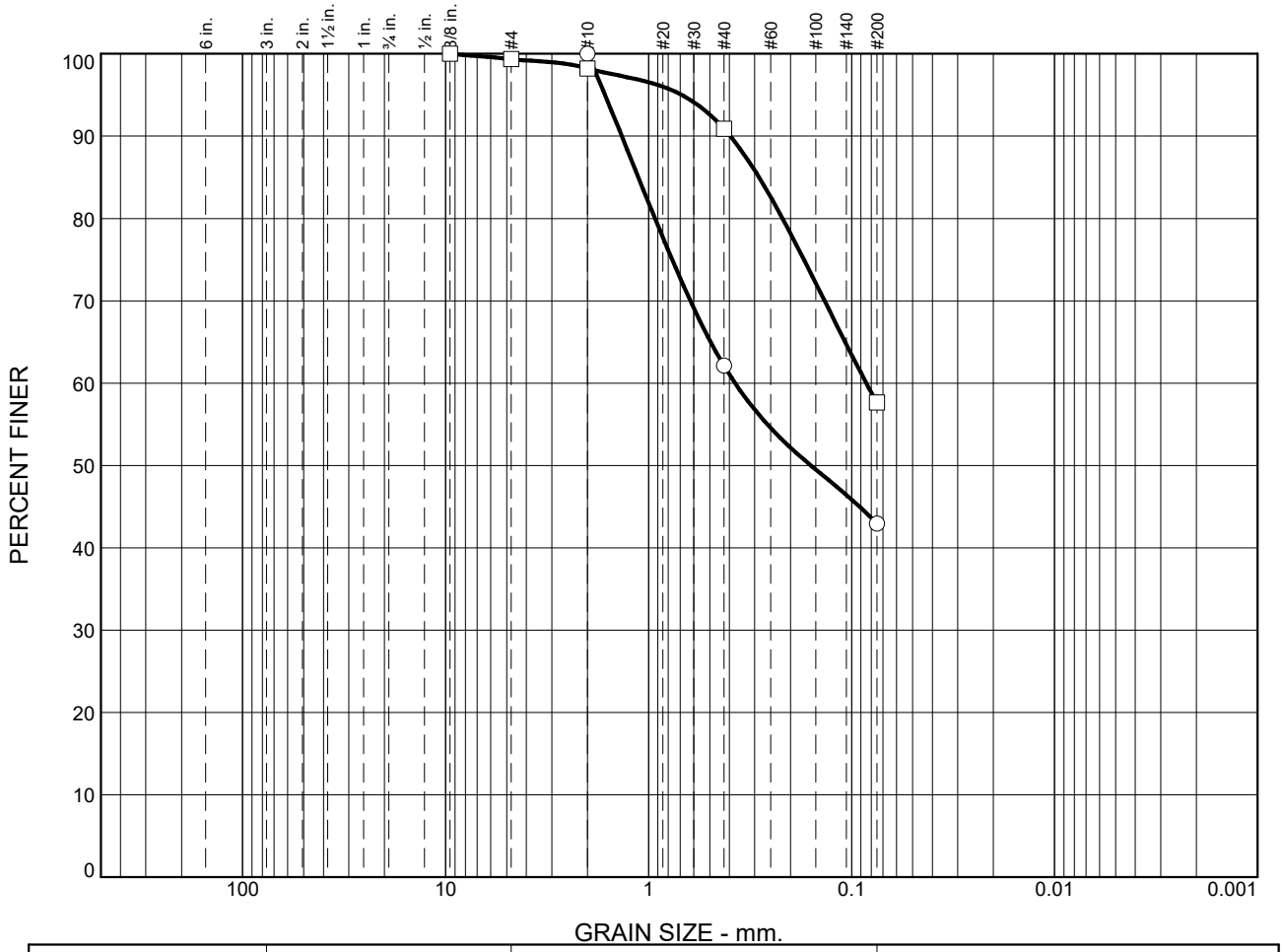
Project: Winding Waters Trail

Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0233-0237

Tested By: CJ/CH **Checked By:** _____

These results are for the exclusive use of the client for whom they were obtained. They apply only to the samples tested and are not indicitive of apparently identical samples.

Particle Size Distribution Report



	% +3"	% Gravel		% Sand			% Fines	
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
○	0.0	0.0	0.0	0.0	37.9	19.1	43.0	
□	0.0	0.0	0.7	1.1	7.3	33.3	57.6	

SOIL DATA

	SOURCE	SAMPLE NO.	DEPTH (ft.)	Material Description	AASHTO
○	W Waters	05/J-4	8-10	silty sand	A-4(0)
□	W Waters	05/J-6	13-15	sandy silt	A-4(0)

FHWA

Sevierville, TN

Client: FHWA/EFLHD

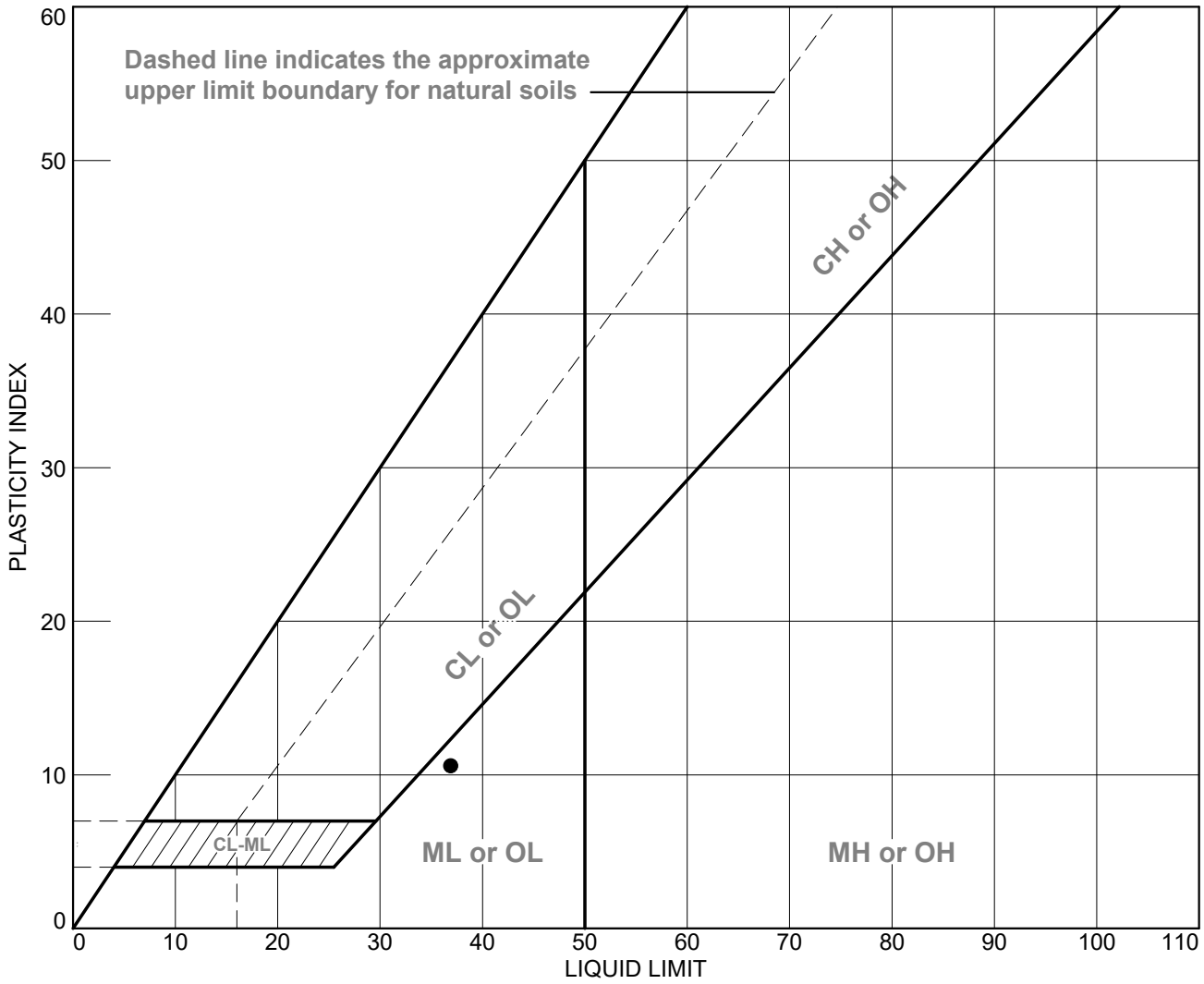
Project: Winding Waters Trail

Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0238-0239

Tested By: CJ/CH **Checked By:** _____

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LIQUID AND PLASTIC LIMITS TEST REPORT



SOIL DATA

SYMBOL	SOURCE	SAMPLE NO.	DEPTH	NATURAL WATER CONTENT (%)	PLASTIC LIMIT (%)	LIQUID LIMIT (%)	PLASTICITY INDEX (%)	USCS
●	W Waters	01/J-3	5-7	34.4	26.3	36.9	10.6	ML
■	W Waters	01/J-5	10-12	21.6	NP	21.6	NP	SM
▲	W Waters	01/J-6	13-15	22.1	NP	NV	NP	ML
◆	W Waters	02/J-4	8-10	87.0	NP	NV	NP	SM
▼	W Waters	02/J-7	15-17	23.8	NP	28.8	NP	ML

FHWA

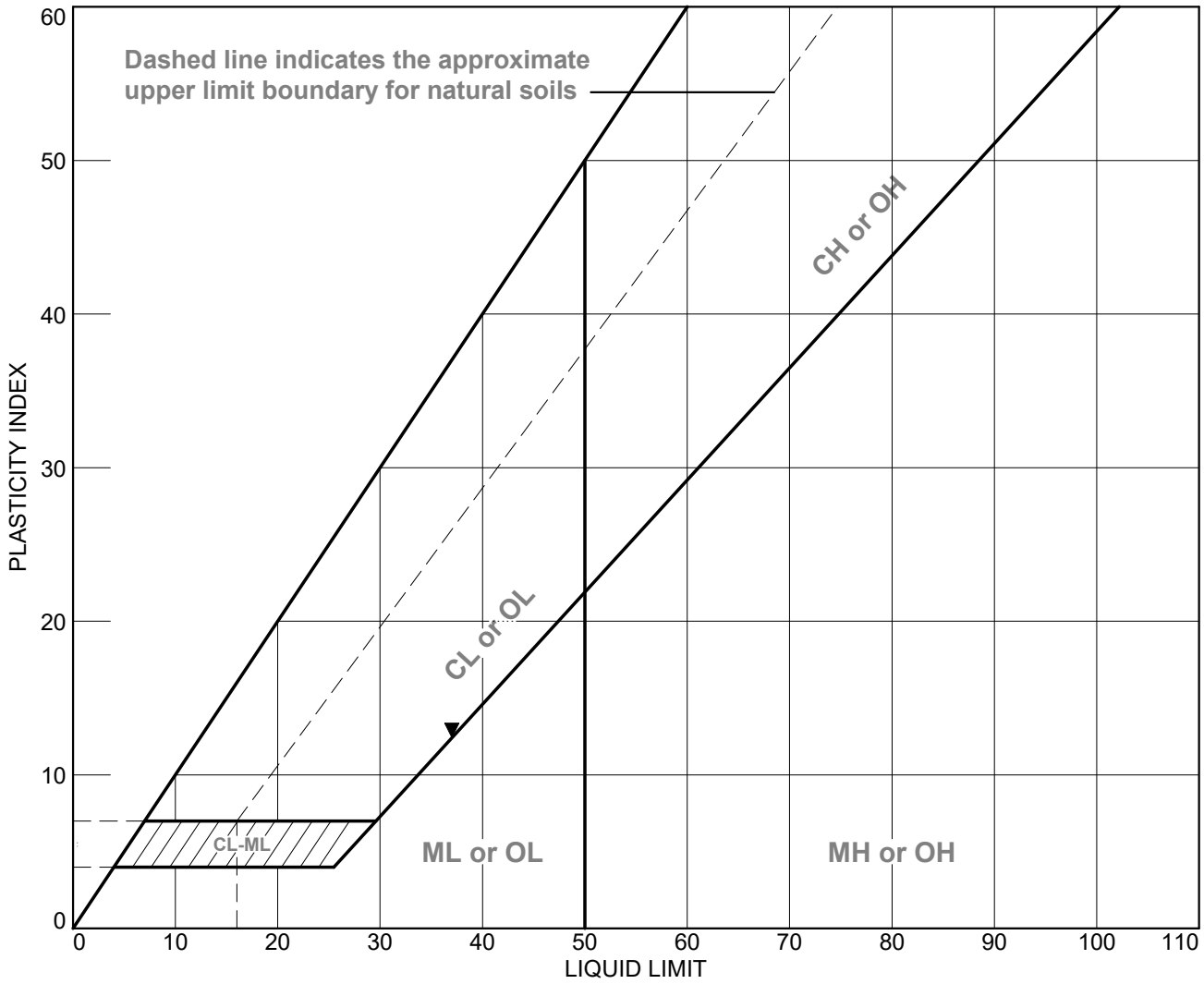
Sevierville, TN

Client: FHWA/EFLHD
Project: Winding Waters Trail
Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0228-0232

Tested By: _____ **Checked By:** _____

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LIQUID AND PLASTIC LIMITS TEST REPORT



SOIL DATA

SYMBOL	SOURCE	SAMPLE NO.	DEPTH	NATURAL WATER CONTENT (%)	PLASTIC LIMIT (%)	LIQUID LIMIT (%)	PLASTICITY INDEX (%)	USCS
●	W Waters	03/J-2/J-3	3-7	28.7	NP	NV	NP	SM
■	W Waters	03/J-4/J-5	8-12	17.0	NP	NV	NP	SM
▲	W Waters	04/J-2/J-3	3-7	35.1	NP	NV	NP	SM
◆	W Waters	04/J-4	8-12	62.9	NP	NV	NP	SM
▼	W Waters	05/J-2/J-3	3-7	37.0	24.1	37.0	12.9	CL

FHWA

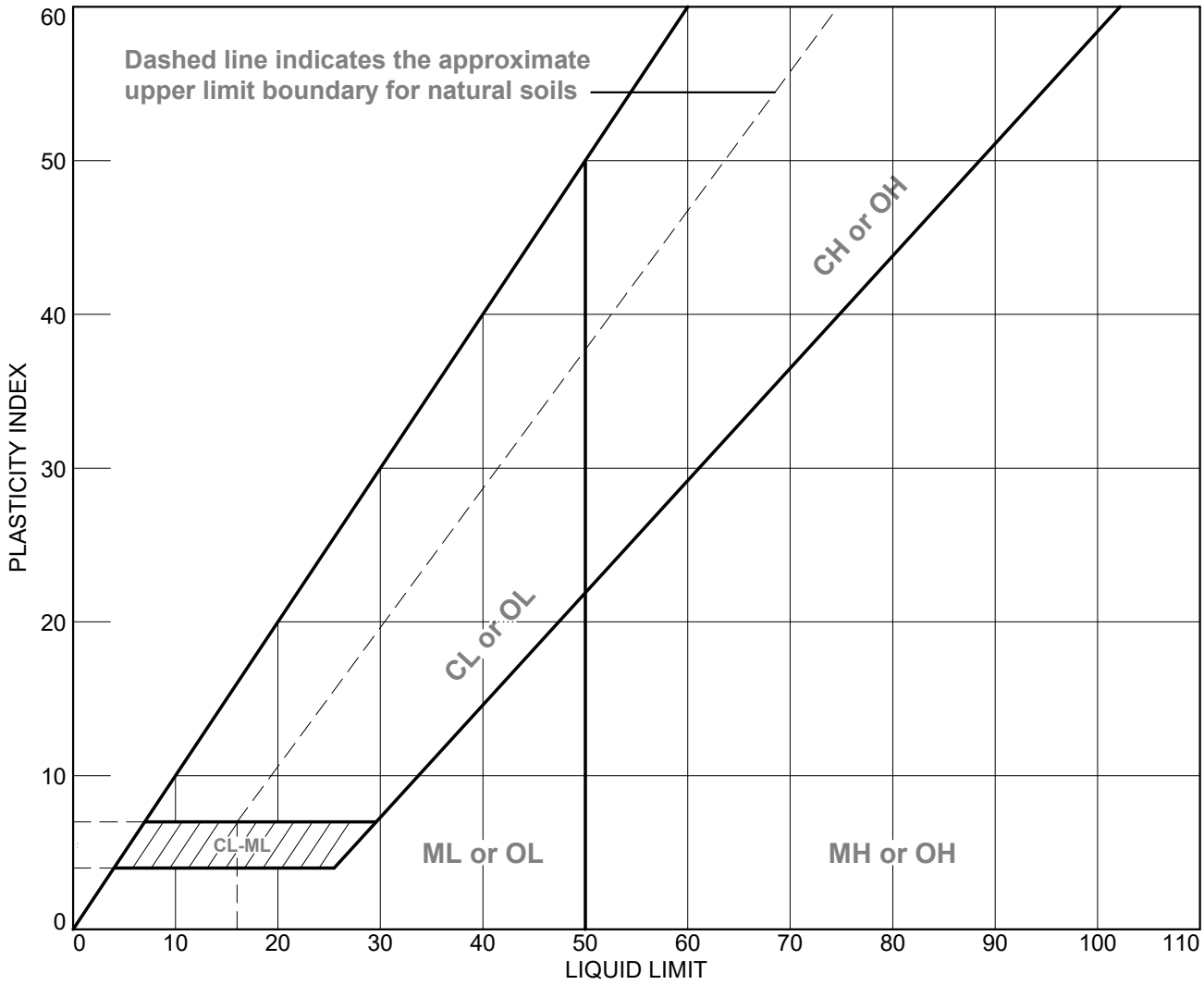
Sevierville, TN

Client: FHWA/EFLHD
Project: Winding Waters Trail
Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0233-0237

Tested By: _____ **Checked By:** _____

These results are for the exclusive use of the client for whom they were obtained. They apply only to the samples tested and are not indicitive of apparently identical samples.

LIQUID AND PLASTIC LIMITS TEST REPORT



SOIL DATA

SYMBOL	SOURCE	SAMPLE NO.	DEPTH	NATURAL WATER CONTENT (%)	PLASTIC LIMIT (%)	LIQUID LIMIT (%)	PLASTICITY INDEX (%)	USCS
●	W Waters	05/J-4	8-10	129.2	NP	NV	NP	SM
■	W Waters	05/J-6	13-15	18.7	NP	NV	NP	ML

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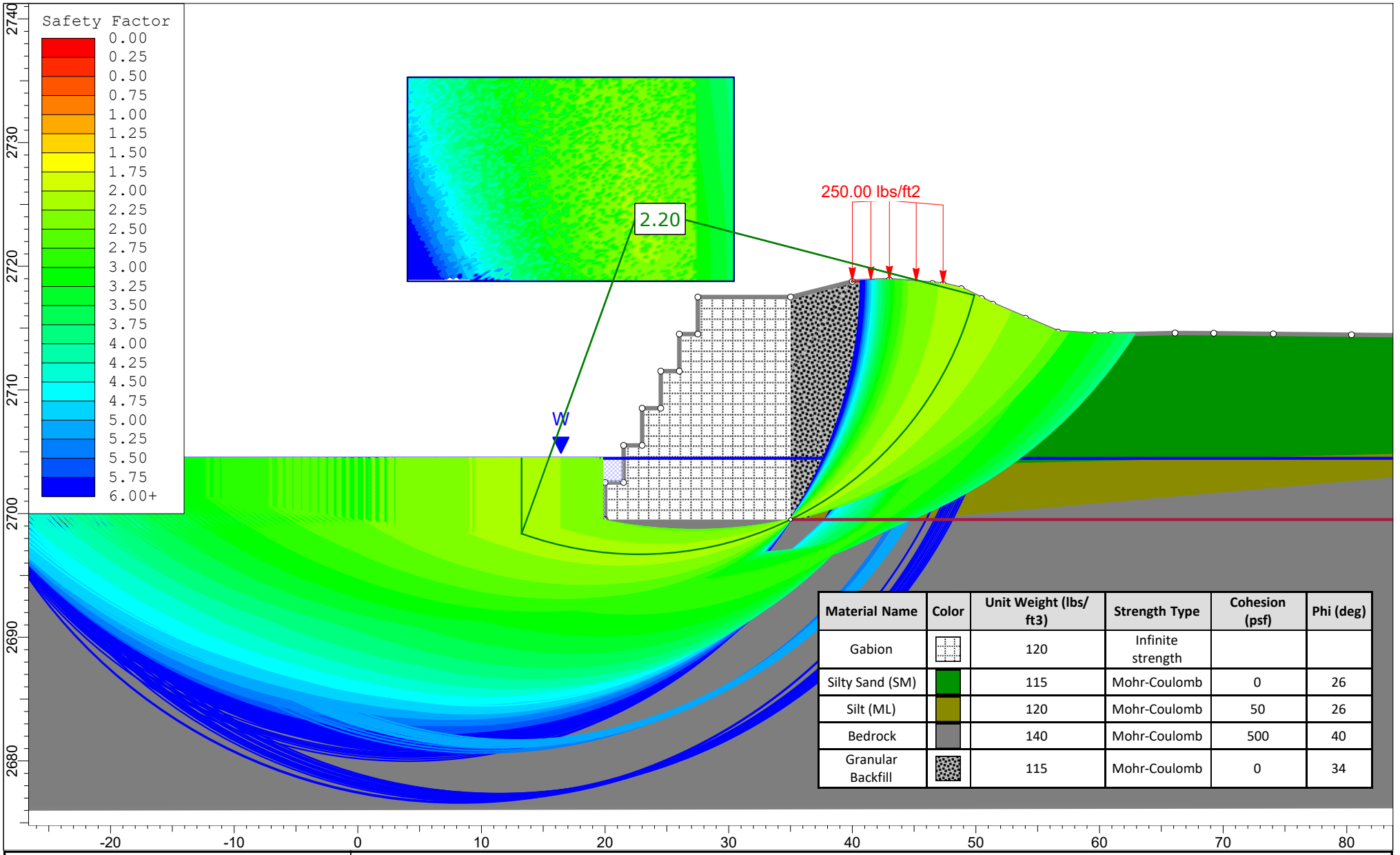
FHWA

Sevierville, TN

Client: FHWA/EFLHD
Project: Winding Waters Trail
Project No.: NY ERFO FW WARI 2019-1(1) **Figure** 0238-0239

Tested By: _____ **Checked By:** _____

Appendix E – Design Analysis

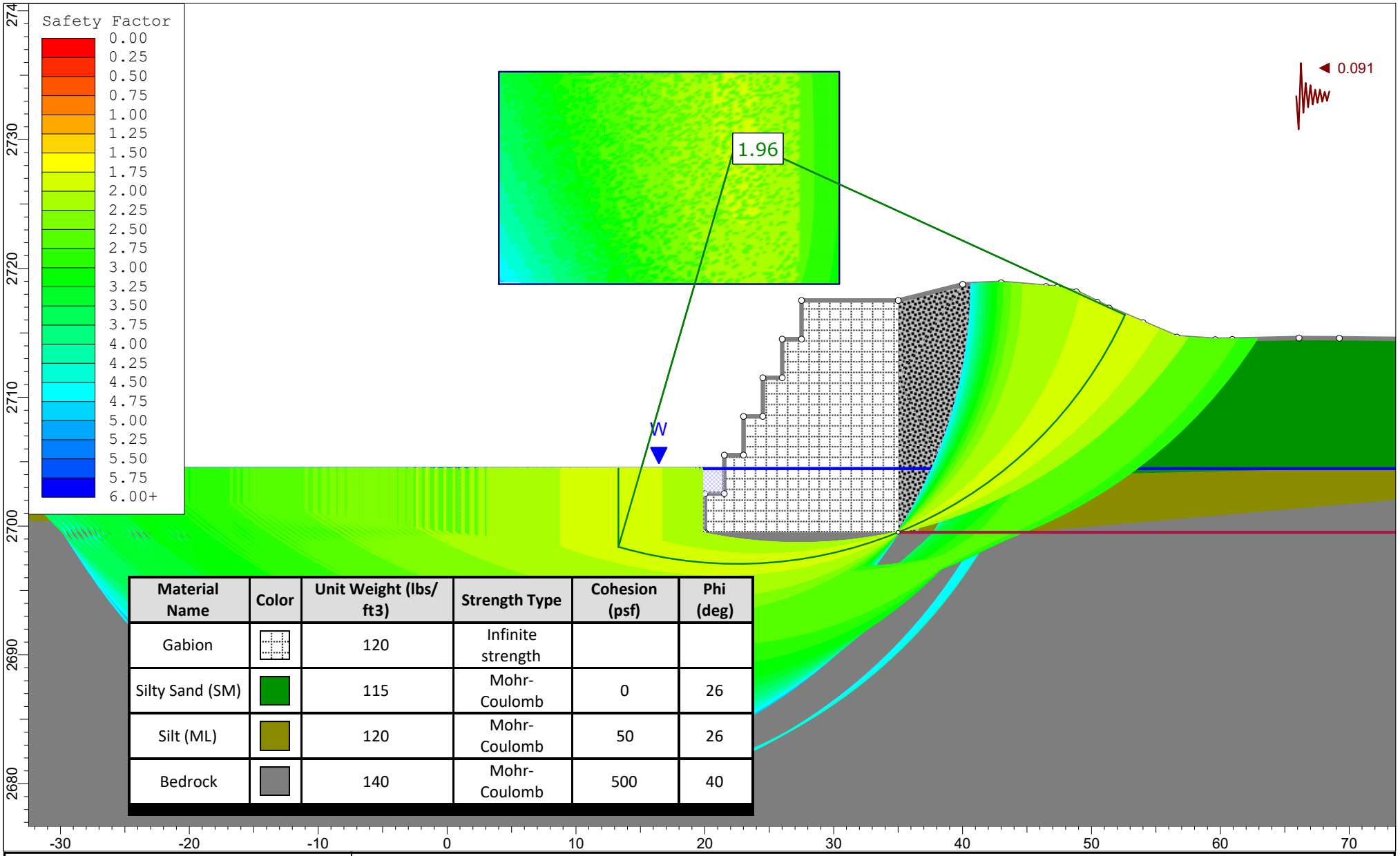


Material Name	Color	Unit Weight (lbs/ft3)	Strength Type	Cohesion (psf)	Phi (deg)
Gabion		120	Infinite strength		
Silty Sand (SM)		115	Mohr-Coulomb	0	26
Silt (ML)		120	Mohr-Coulomb	50	26
Bedrock		140	Mohr-Coulomb	500	40
Granular Backfill		115	Mohr-Coulomb	0	34

U.S. Department of Transportation
Federal Highway Administration

SLIDEINTERPRET 9.001

Project	NY ERFO FW WARI 2019-1(1)		
Group	Water Table at 373.5ft	*Scenario*	Global
Drawn By	YPS	*Company*	
Date	12/20/2022, 10:30:59 AM	*File Name*	SlideModel.slmd

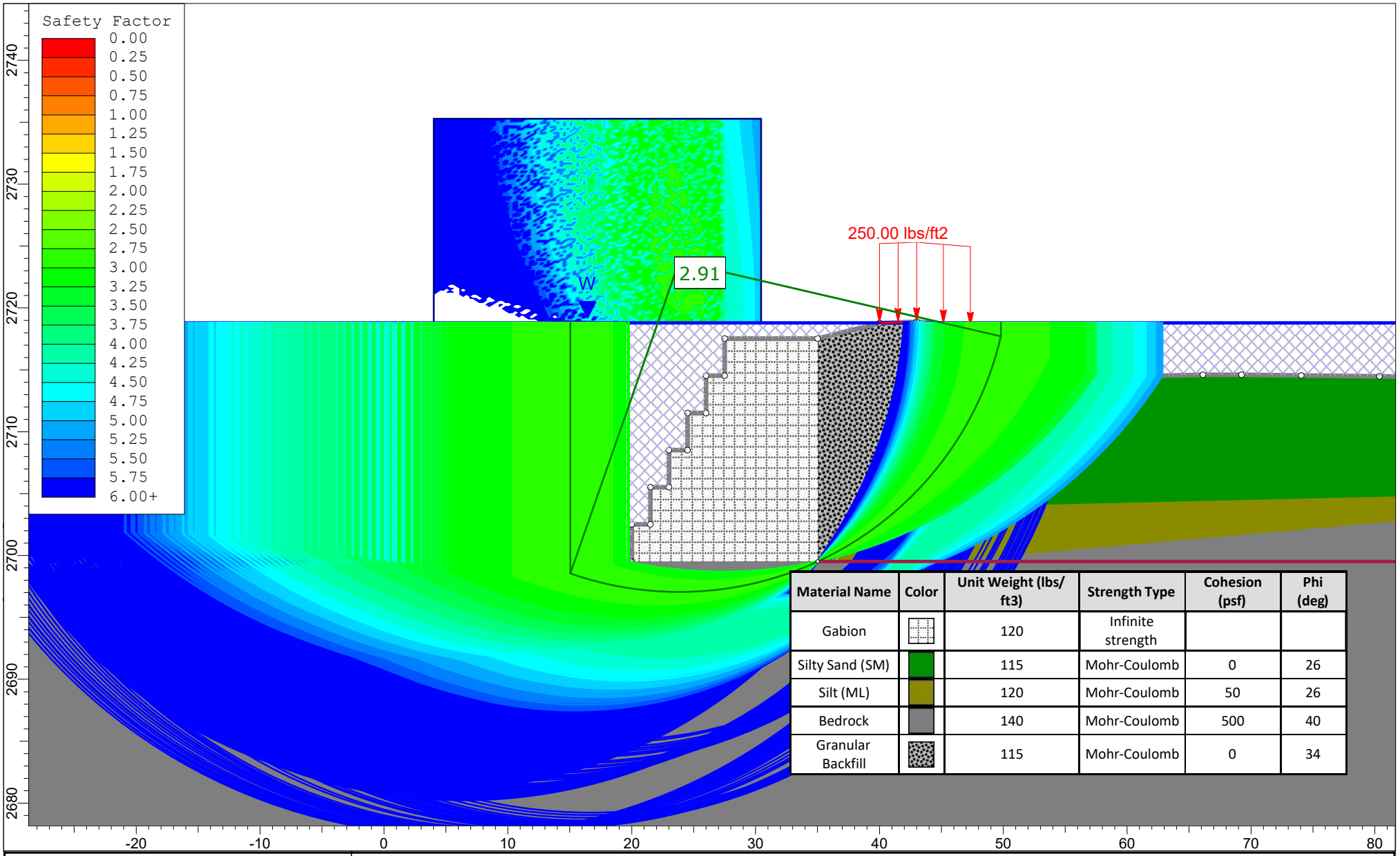


Material Name	Color	Unit Weight (lbs/ ft3)	Strength Type	Cohesion (psf)	Phi (deg)
Gabion		120	Infinite strength		
Silty Sand (SM)		115	Mohr-Coulomb	0	26
Silt (ML)		120	Mohr-Coulomb	50	26
Bedrock		140	Mohr-Coulomb	500	40

U.S. Department of Transportation
Federal Highway Administration

SLIDEINTERPRET 9.001

Project	NY ERFO FW WARI 2019-1(1)		
Group	Water Table at 373.5ft	*Scenario*	Seismic
Drawn By	YPS	*Company*	
Date	12/20/2022, 10:30:59 AM	*File Name*	SlideModel.slmd



Material Name	Color	Unit Weight (lbs/ ft3)	Strength Type	Cohesion (psf)	Phi (deg)
Gabion		120	Infinite strength		
Silty Sand (SM)		115	Mohr-Coulomb	0	26
Silt (ML)		120	Mohr-Coulomb	50	26
Bedrock		140	Mohr-Coulomb	500	40
Granular Backfill		115	Mohr-Coulomb	0	34

U.S. Department of Transportation
Federal Highway Administration

SLIDEINTERPRET 9.001

Project	NY ERFO FW WARI 2019-1(1)		
Group	Extreme Event II - 100year	*Scenario*	Global
Drawn By	YPS	*Company*	
Date	12/20/2022, 10:30:59 AM	*File Name*	SlideModel.slmd

Appendix F – Site Photographs



Photo 1: Site 240 looking East.



Photo 2: Set up of boring BH22-01.



Photo 3: Set up of boring BH22-02.



Photo 4: Set up of boring BH22-03.



Photo 5: Site 324, looking Northeast.



Photo 6: Set up of boring BH22-04.



Photo 7: : Set up of boring BH22-05.